

Math 1352 Exam II

Discussion Time and TA:

Name:

*Calculators are not allowed on this exam. Work all questions completely. Show all work as described in class.
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1. Evaluate the following integrals.

3 (a) $\int \csc x \, dx$

3 (b) $\int \frac{1}{2\sqrt{x}} \, dx$

3 (c) $\int \frac{1}{\sqrt{1-x^2}} \, dx$

3 (d) $\int \tan x \, dx$

3 (e) $\int \cosh x \, dx$

3 (f) $\int \frac{1}{x} \, dx$

10 (g) $\int \frac{3x}{(x-2)^2} \, dx$

8 (h) $\int \sinh x \cosh^3 x \, dx$

10 (i) $\int x^2 e^{3x} \, dx$

8 (j) $\int \frac{2x}{x^2 - 3} dx$

10 (k) $\int \frac{1}{\sqrt{x^2 + 9}} dx$

10 2. Set up the partial fraction decomposition for $\frac{7x^2 - x}{x(x - 3)^2(x^2 + 4x + 7)^2}$.

3. Do the following sequences converge? If so, find the limit; if not, explain why not.

4 (a) $\left\{ \frac{2n^3 - 4}{3n^4 + 5n} \right\}$

4 (b) $\left\{ \frac{n + 7}{\ln(n) - 4} \right\}$

10 4. Does the integral $\int_e^\infty \frac{1}{x(\ln x)^3} dx$ converge? If so, evaluate the integral. If not, explain why.

8 5. Indicate if the following are always true or may be false.

(a) Every bounded sequence converges.

(b) Every strictly increasing sequence which is bounded above converges.

(c) Every decreasing sequence is monotone.

(d) The sequence $\left\{ \left(1 + \frac{1}{n} \right)^n \right\}$ converges to 1.